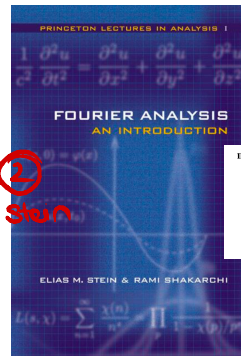
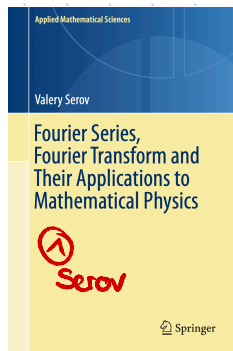


Kapitel 2: Fourier Transform

1. Grundlagen
2. Konvergenz
3. Deconvolving
4. Limitations



1. Theoretical Foundations

1.1. Fourier-Reihe

1.1 Main definitions and some examples

We now begin our study of Fourier analysis with the precise definition of the Fourier series of a function. Here, it is important to pin down where our function is originally defined. If f is an integrable function given on an interval $[a, b]$ of length L (that is, $b - a = L$), then the n^{th} **Fourier coefficient** of f is defined by

$$\hat{f}(n) = \frac{1}{L} \int_a^b f(x) e^{-2\pi i n x / L} dx, \quad n \in \mathbb{Z}.$$

The **Fourier series** of f is given formally³ by

$$\sum_{n=-\infty}^{\infty} \hat{f}(n) e^{2\pi i n x / L}.$$

We shall sometimes write a_n for the Fourier coefficients of f , and use the notation

$$f(x) \sim \sum_{n=-\infty}^{\infty} a_n e^{2\pi i n x / L}$$

to indicate that the series on the right-hand side is the Fourier series of f .

For instance, if f is an integrable function on the interval $[-\pi, \pi]$, then the n^{th} Fourier coefficient of f is

$$\hat{f}(n) = a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) e^{-in\theta} d\theta, \quad n \in \mathbb{Z},$$

and the Fourier series of f is

$$f(\theta) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}.$$

Here we use θ as a variable since we think of it as an angle ranging from $-\pi$ to π .

³At this point, we do not say anything about the convergence of the series.

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1.2 Definition of the Fourier Transform

We shall denote by $\mathcal{M}(\mathbb{R})$ the set of functions of moderate decrease on $\mathbb{R}$. As an exercise, the reader can check that under the usual addition of functions and multiplication by scalars, $\mathcal{M}(\mathbb{R})$ forms a vector space over $\mathbb{C}$.

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1.2 Definition of the Fourier transform

If $f \in \mathcal{M}(\mathbb{R})$, we define its **Fourier transform** for $\xi \in \mathbb{R}$ by

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx.$$

Of course, $|e^{-2\pi i x \xi}| = 1$, so the integrand is of moderate decrease, and the integral makes sense.

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In fact, this last observation implies that $\hat{f}$ is bounded, and moreover, a simple argument shows that $\hat{f}$ is continuous and tends to 0 as $|\xi| \rightarrow \infty$ (Exercise 5). However, nothing in the definition above guarantees that $\hat{f}$ is of moderate decrease, or has a specific decay. In particular, it is not clear in this context how to make sense of the integral $\int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi$ and the resulting Fourier inversion formula. To remedy this, we introduce a more refined space of functions considered by Schwartz which is very

Let us consider a series of the form

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right). \quad (2.1)$$

This series consists of $2L$ -periodic functions. Thus, if the series (2.1) converges for all x , then the function to which it converges will also be $2L$ -periodic. Let us denote this limiting function by $f(x)$, i.e.,

$$f(x) := \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right). \quad (2.2)$$

To determine a_n and b_n we proceed as follows: assuming that the integration can be legitimately carried out term by term (it will be, for example, if $\sum_{n=1}^{\infty} (|a_n| + |b_n|) < \infty$), we obtain

$$\begin{aligned} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx &= \frac{a_0}{2} \int_{-L}^L \cos \frac{n\pi x}{L} dx + \sum_{m=1}^{\infty} a_m \int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx \\ &\quad + \sum_{m=1}^{\infty} b_m \int_{-L}^L \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx \end{aligned}$$

for each fixed $n = 1, 2, \dots$. It follows from the orthogonality relations (1.10), (1.11), and (1.13) that the only nonzero term on the right-hand side is the one for which $m = n$ in the first summation. Hence

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad n = 1, 2, \dots \quad (2.3)$$

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DOI 10.1007/978-3-319-65262-7_2

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Part I: Fourier Series and the Discrete Fourier Transform

A similar expression for b_n is obtained by multiplying (2.2) by $\sin \frac{n\pi x}{L}$ and integrating termwise from $-L$ to L . The result is

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx, \quad n = 1, 2, \dots \quad (2.4)$$

Using (1.13) we can easily obtain that

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx. \quad (2.5)$$

Definition 2.1. Let f be integrable (not necessarily periodic) on the interval $[-L, L]$. The **Fourier series** of f is the **trigonometric series** (2.1), where the coefficients a_0, a_n and b_n are given by (2.5), (2.3), and (2.4), respectively. In that case, we write

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right). \quad (2.6)$$

Remark 2.2. This definition does not imply that the series (2.6) converges to f or that f is periodic.

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These three alternatives allow us to consider (for simplicity) only the case of a symmetric interval $[-\pi, \pi]$ such that the Fourier series will be of the form (2.9) i.e.

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$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx).$$

Using Euler's formula, we will rewrite this series in the complex form

$$f(x) \sim \sum_{n=-\infty}^{\infty} c_n e^{inx}, \quad (2.10)$$

where the coefficients $c_n = c_n(f)$ are equal to

$$c_n = \begin{cases} \frac{a_0}{2}, & n = 0, \\ \frac{a_n - ib_n}{2}, & n = 1, 2, \dots, \\ \frac{a_n + ib_n}{2}, & n = -1, -2, \dots \end{cases} \quad \begin{cases} a_n = c_n + c_{-n}, & n = 1, 2, \dots, \\ a_0 = 2c_0, & \\ b_n = i(c_n - c_{-n}), & n = 1, 2, \dots \end{cases} \quad (2.11)$$

The formulas (2.3), (2.4), (2.5), and (2.11) imply that

$$c_n(f) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \quad (2.12)$$

for $n = 0, \pm 1, \pm 2, \dots$. We call $c_n(f)$ the n th Fourier coefficient of f . It can be checked that

$$c_n(f) = c_{-n}(\bar{f}). \quad (2.13)$$

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useful in establishing the initial properties of the Fourier transform. The choice of the Schwartz space is motivated by an important principle which ties the decay of $\hat{f}$ to the continuity and differentiability properties of f (and vice versa): the faster $\hat{f}(\xi)$ decreases as $|\xi| \rightarrow \infty$, the “smoother” f must be. An example that reflects this principle is given in Exercise 3. We also note that this relationship between f and $\hat{f}$ is reminiscent of a similar one between the smoothness of a function on the circle and the decay of its Fourier coefficients; see the discussion of Corollary 2.4 in Chapter 2.

1.3 The Schwartz space

The **Schwartz space** on $\mathbb{R}$ consists of the set of all indefinitely differentiable functions f so that f and all its derivatives $f', f'', \dots, f^{(\ell)}, \dots$,

$\Leftarrow$

Definition 16.1. The *Schwartz space* $S(\mathbb{R}^n)$ of rapidly decaying functions is defined as

$$S(\mathbb{R}^n) = \{f \in C^\infty(\mathbb{R}^n) : |f|_{\alpha, \beta} := \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta f(x)| < \infty \text{ for any } \alpha, \beta \in \mathbb{N}_0^n\}.$$

The following properties of $S = S(\mathbb{R}^n)$ are readily verified.

- (1) S is a linear space.
- (2) $\partial^\alpha : S \rightarrow S$ for every $\alpha \geq 0$.
- (3) $x^\beta : S \rightarrow S$ for every $\beta \geq 0$.
- (4) If $f \in S(\mathbb{R}^n)$, then $|f(x)| \leq c_m(1 + |x|)^{-m}$ for every $m \in \mathbb{N}$. The converse is not true (see part (3) of Example 16.2).
- (5) It follows from part (4) that $S(\mathbb{R}^n) \subset L^p(\mathbb{R}^n)$ for every $1 \leq p \leq \infty$.

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1. Elementary theory of the Fourier transform

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are **rapidly decreasing**, in the sense that

$$\sup_{x \in \mathbb{R}} |x|^k |f^{(\ell)}(x)| < \infty \quad \text{for every } k, \ell \geq 0.$$

We denote this space by $\mathcal{S} = \mathcal{S}(\mathbb{R})$, and again, the reader should verify that $\mathcal{S}(\mathbb{R})$ is a vector space over $\mathbb{C}$. Moreover, if $f \in \mathcal{S}(\mathbb{R})$, we have

$$f'(x) = \frac{df}{dx} \in \mathcal{S}(\mathbb{R}) \quad \text{and} \quad xf(x) \in \mathcal{S}(\mathbb{R}).$$

This expresses the important fact that the Schwartz space is closed under differentiation and multiplication by polynomials.

A simple example of a function in $\mathcal{S}(\mathbb{R})$ is the **Gaussian** defined by

$$f(x) = e^{-x^2},$$

which plays a central role in the theory of the Fourier transform, as well as other fields (for example, probability theory and physics). The reader can check that the derivatives of f are of the form $P(x)e^{-x^2}$ where P is a polynomial, and this immediately shows that $f \in \mathcal{S}(\mathbb{R})$. In fact, e^{-ax^2} belongs to $\mathcal{S}(\mathbb{R})$ whenever $a > 0$. Later, we will normalize the Gaussian by choosing $a = \pi$.

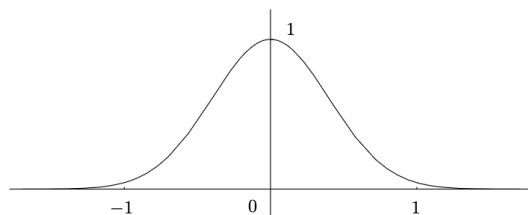


Figure 1. The Gaussian $e^{-\pi x^2}$

An important class of other examples in $\mathcal{S}(\mathbb{R})$ are the “bump functions” which vanish outside bounded intervals (Exercise 4).

As a final remark, note that although $e^{-|x|}$ decreases rapidly at infinity, it is not differentiable at 0 and therefore does not belong to $\mathcal{S}(\mathbb{R})$.

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1.4 The Fourier transform on $\mathcal{S}$

The **Fourier transform** of a function $f \in \mathcal{S}(\mathbb{R})$ is defined by

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx.$$

Some simple properties of the Fourier transform are gathered in the following proposition. We use the notation

$$f(x) \longrightarrow \hat{f}(\xi)$$

to mean that $\hat{f}$ denotes the Fourier transform of f .

Proposition 1.2 If $f \in \mathcal{S}(\mathbb{R})$ then:

- (i) $f(x + h) \longrightarrow \hat{f}(\xi) e^{2\pi i h \xi}$ whenever $h \in \mathbb{R}$.
- (ii) $f(x) e^{-2\pi i x h} \longrightarrow \hat{f}(\xi + h)$ whenever $h \in \mathbb{R}$.
- (iii) $f(\delta x) \longrightarrow \delta^{-1} \hat{f}(\delta^{-1} \xi)$ whenever $\delta > 0$.
- (iv) $f'(x) \longrightarrow 2\pi i \xi \hat{f}(\xi)$.
- (v) $-2\pi i x f(x) \longrightarrow \frac{d}{d\xi} \hat{f}(\xi)$.

In particular, except for factors of $2\pi i$, the Fourier transform interchanges differentiation and multiplication by x . This is the key property that makes the Fourier transform a central object in the theory of differential equations. We shall return to this point later.

Definition 16.5. The *Fourier transform* $\mathcal{F}f(\xi)$ or $\hat{f}(\xi)$ of the function $f(x) \in \mathcal{S}$ is defined by

$$\mathcal{F}f(\xi) \equiv \hat{f}(\xi) := (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-i(x, \xi)} f(x) dx, \quad \xi \in \mathbb{R}^n.$$

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1.3. Eigenschaften / Properties der FT

Next we prove the following properties of the Fourier transform:

- (1) $\mathcal{F}$ is a linear continuous map from S into S . **LINEARITÄT**
 (2) $\xi^\alpha \partial_\xi^\beta \hat{f}(\xi) = (-i)^{|\alpha|+|\beta|} \partial_x^\alpha (x^\beta f(x))$.

Indeed, we have

$$\partial_\xi^\beta \hat{f}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} (-ix)^\beta e^{-i(x,\xi)} f(x) dx$$

and hence

$$\|\partial_\xi^\beta \hat{f}(\xi)\|_{L^\infty(\mathbb{R}^n)} \leq c_m (2\pi)^{-n/2} \int_{\mathbb{R}^n} \frac{|x|^{|\beta|}}{(1+|x|)^m} dx < \infty$$

if we choose $m > n + |\beta|$. At the same time we have obtained the formula

$$\partial_\xi^\beta \hat{f}(\xi) = \widehat{(-ix)^\beta f(x)}. \quad (16.1)$$

Further, integration by parts gives us

$$\xi^\alpha \hat{f}(\xi) = (-i)^{|\alpha|} (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-i(x,\xi)} \partial_x^\alpha f(x) dx,$$

from which we have the estimate

$$\|\xi^\alpha \hat{f}\|_{L^\infty(\mathbb{R}^n)} \leq c \int_{\mathbb{R}^n} |\partial_x^\alpha f(x)| dx < \infty,$$

since $\partial_x^\alpha f(x) \in S$ for every $\alpha \geq 0$ if $f(x) \in S$. And also we have the formula

$$\xi^\alpha \hat{f} = (-i)^{|\alpha|} \widehat{\partial_x^\alpha f}. \quad (16.2)$$

If we combine these last two estimates, we may conclude that $\mathcal{F} : S \rightarrow S$ and $\mathcal{F}$ is a continuous map (in the sense of the metric space (S, ρ)), since $\mathcal{F}$ maps every bounded set from S again to a bounded set from S .

The formulas (16.1) and (16.2) show us that it is more convenient to use the following notation:

$$D_j = -i\partial_j = -i \frac{\partial}{\partial x_j}, \quad D^\alpha = D_1^{\alpha_1} \dots D_n^{\alpha_n}.$$

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Definition 16.8. We adopt the following notation for *translation* and *dilation* of a function:

$$(t_h f)(x) := f(x-h), \quad (\sigma_\lambda f)(x) := f(\lambda x), \quad \lambda \neq 0.$$

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1.4. Inverse FT

1.5 The Fourier inversion

The next result is an identity sometimes called the multiplication formula.

Proposition 1.8 If $f, g \in S(\mathbb{R})$, then

$$\int_{-\infty}^{\infty} f(x) \hat{g}(x) dx = \int_{-\infty}^{\infty} \hat{f}(y) g(y) dy.$$

To prove the proposition, we need to digress briefly to discuss the interchange of the order of integration for double integrals. Suppose $F(x, y)$ is a continuous function in the plane $(x, y) \in \mathbb{R}^2$. We will assume the following decay condition on F :

$$|F(x, y)| \leq A/(1+x^2)(1+y^2).$$

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Then, we can state that for each x the function $F(x, y)$ is of moderate decrease in y , and similarly for each fixed y the function $F(x, y)$ is of moderate decrease in x . Moreover, the function $F_1(x) = \int_{-\infty}^{\infty} F(x, y) dy$ is continuous and of moderate decrease; similarly for the function $F_2(y) = \int_{-\infty}^{\infty} F(x, y) dx$. Finally

$$\int_{-\infty}^{\infty} F_1(x) dx = \int_{-\infty}^{\infty} F_2(y) dy.$$

The proof of these facts may be found in the appendix.

We now apply this to $F(x, y) = f(x)g(y)e^{-2\pi ixy}$. Then $F_1(x) = f(x)\hat{g}(x)$, and $F_2(y) = \hat{f}(y)g(y)$ so

$$\int_{-\infty}^{\infty} f(x)\hat{g}(x) dx = \int_{-\infty}^{\infty} \hat{f}(y)g(y) dy,$$

which is the assertion of the proposition.

The multiplication formula and the fact that the Gaussian is its own Fourier transform lead to a proof of the first major theorem.

Theorem 1.9 (Fourier inversion) If $f \in S(\mathbb{R})$, then

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi.$$

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Corollary 1.10 The Fourier transform is a bijective mapping on the Schwartz space.

1.6 The Plancherel formula

We need a few further results about convolutions of Schwartz functions. The key fact is that the Fourier transform interchanges convolutions with pointwise products, a result analogous to the situation for Fourier series.

Proposition 1.11 If $f, g \in S(\mathbb{R})$ then:

- (i) $f * g \in S(\mathbb{R})$.

V + Faltungssatz

Remark 16.9. Here $\mathcal{F}^*$ is the adjoint operator (in the sense of L^2), which maps S into S since $\mathcal{F} : S \rightarrow S$. The inverse Fourier transform $\mathcal{F}^{-1}$ is defined as $\mathcal{F}^{-1} := \mathcal{F}^*$.

In order to justify this definition we will prove the following theorem.

Theorem 16.10 (Fourier inversion formula). Let f be a function from $S(\mathbb{R}^n)$. Then

$$\mathcal{F}^* \mathcal{F} f = f.$$

To this end we will prove first the following (somewhat technical) lemma.

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(ii) $f * g = g * f$.

(iii) $(\widehat{f * g})(\xi) = \widehat{f}(\xi)\widehat{g}(\xi)$.

Proof. To prove that $f * g$ is rapidly decreasing, observe first that for any $\ell \geq 0$ we have $\sup_x |x|^\ell |g(x-y)| \leq A_\ell(1+|y|)^\ell$, because g is rapidly decreasing (to check this assertion, consider separately the two cases $|x| \leq 2|y|$ and $|x| \geq 2|y|$). From this, we see that

$$\sup_x |x^\ell (f * g)(x)| \leq A_\ell \int_{-\infty}^{\infty} |f(y)|(1+|y|)^\ell dy,$$

so that $x^\ell (f * g)(x)$ is a bounded function for every $\ell \geq 0$. These estimates carry over to the derivatives of $f * g$, thereby proving that $f * g \in \mathcal{S}(\mathbb{R})$ because

$$\left(\frac{d}{dx}\right)^k (f * g)(x) = (f * \left(\frac{d}{dx}\right)^k g)(x) \quad \text{for } k = 1, 2, \dots$$

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This identity is proved first for $k = 1$ by differentiating under the integral defining $f * g$. The interchange of differentiation and integration is justified in this case by the rapid decrease of dg/dx . The identity then follows for every k by iteration.

For fixed x , the change of variables $x - y = u$ shows that

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x - u)g(u) du = (g * f)(x).$$

This change of variables is a composition of two changes, $y \mapsto -y$ and $y \mapsto y - h$ (with $h = x$). For the first one we use the observation that $\int_{-\infty}^{\infty} F(x) dx = \int_{-\infty}^{\infty} F(-x) dx$ for any Schwartz function F , and for the second, we apply (ii) of Proposition 1.1

Finally, consider $F(x, y) = f(y)g(x - y)e^{-2\pi i x \xi}$. Since f and g are rapidly decreasing, considering separately the two cases $|x| \leq 2|y|$ and $|x| \geq 2|y|$, we see that the discussion of the change of order of integration after Proposition 1.8 applies to F . In this case $F_1(x) = (f * g)(x)e^{-2\pi i x \xi}$, and $F_2(y) = f(y)e^{-2\pi i y \xi} \widehat{g}(\xi)$. Thus $\int_{-\infty}^{\infty} F_1(x) dx = \int_{-\infty}^{\infty} F_2(y) dy$, which implies (iii). The proposition is therefore proved.

We now use the properties of convolutions of Schwartz functions to prove the main result of this section. The result we have in mind is the analogue for functions on $\mathbb{R}$ of Parseval's identity for Fourier series.

The Schwartz space can be equipped with a Hermitian inner product

$$(f, g) = \int_{-\infty}^{\infty} f(x)\overline{g(x)} dx$$

whose associated norm is

$$\|f\| = \left(\int_{-\infty}^{\infty} |f(x)|^2 dx \right)^{1/2}.$$

The second major theorem in the theory states that the Fourier transform is a unitary transformation on $\mathcal{S}(\mathbb{R})$.

Theorem 1.12 (Plancherel) *If $f \in \mathcal{S}(\mathbb{R})$ then $\|\widehat{f}\| = \|f\|$.*

~ dh fr ist unitäre Transformation auf $\mathcal{S}(\mathbb{R})$

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1.7 Extension to functions of moderate decrease

In the previous sections, we have limited our assertion of the Fourier inversion and Plancherel formulas to the case when the function involved belonged to the Schwartz space. It does not really involve further ideas to extend these results to functions of moderate decrease, once we make the additional assumption that the Fourier transform of the function under consideration is also of moderate decrease. Indeed, the key observation, which is easy to prove, is that the convolution $f * g$ of two functions f and g of moderate decrease is again a function of moderate decrease (Exercise 7); also $\widehat{f * g} = \widehat{f}\widehat{g}$. Moreover, the multiplication formula continues to hold, and we deduce the Fourier inversion and Plancherel formulas when f and $\widehat{f}$ are both of moderate decrease.

This generalization, although modest in scope, is nevertheless useful in some circumstances.

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1.5 Discrete FT - DFT

Chapter 11 Formulation of the Discrete Fourier Transform and Its Properties.

Let $x(t)$ be a 2π -periodic continuous signal. Assume that $x(t)$ can be represented by an absolutely convergent trigonometric Fourier series

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jnt}, \quad t \in [-\pi, \pi], \quad (11.1)$$

where the c_n are the Fourier coefficients of $x(t)$.

Let now N be an even positive integer and

$$t_k = \frac{2\pi k}{N}, \quad k = -\frac{N}{2}, \dots, \frac{N}{2} - 1.$$

Then $x(t_k)$ is a response at t_k , i.e.,

$$x(t_k) = \sum_{n=-\infty}^{\infty} c_n e^{jnt_k}. \quad (11.2)$$

Since $e^{j2\pi k l} = 1$ for integers k and l , the series (11.2) can be rewritten as

$$\begin{aligned} \sum_{n=-\infty}^{\infty} c_n e^{jnt_k} &= \sum_{l=-\infty}^{\infty} \sum_{n=-N/2}^{N/2-1} c_n e^{j(n+2\pi k l)t_k} \\ &= \sum_{l=-\infty}^{\infty} \sum_{n=-N/2}^{N/2-1} c_{n+2\pi k l} e^{jnt_k} \\ &= \sum_{n=-N/2}^{N/2-1} c_n e^{jnt_k} + \sum_{n=-N/2}^{N/2-1} c_{n+2\pi k} e^{jnt_k} + \dots \end{aligned} \quad (11.3)$$

where $X_{n,n} = -N/2, \dots, N/2 - 1$ is given by

① S. 77 | Ch. 11

Theorem 17.5 (Plancherel). *Let $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ be the Fourier transform in $\mathcal{S}$ with $\|\mathcal{F}f\|_{L^2} = \|f\|_{L^2}$. Then there exists a unique extension $\mathcal{F}_{\text{ex}}$ of $\mathcal{F}$ to L^2 -space such that $\mathcal{F}_{\text{ex}} : L^2 \xrightarrow{\text{meas}} L^2$ and $\|\mathcal{F}_{\text{ex}}\|_{L^2 \rightarrow L^2} = 1$. Also Parseval's equality remains valid.*

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Remark 17.6. In L^2 -space we also have the Fourier inversion formula $\mathcal{F}_{\text{ex}}^* \mathcal{F}_{\text{ex}} f = f$, or $\mathcal{F}_{\text{ex}}^{-1} \mathcal{F}_{\text{ex}} f = f$.

Inversion gilt auch in L^2

① S. 146 | Ch. 17

Discrete Fourier Series:

$$X(j) = \sum_{n=0}^{N-1} A(n)W_N^{jn} \quad \text{where } W_N = \exp(2\pi j/N)$$

Inversion Formula:

$$A(n) = \frac{1}{N} \sum_{j=0}^{N-1} X(j)W_N^{-jn}$$

Let us write: $X(j) \leftrightarrow A(n)$

Periodicity:

$$\begin{aligned} W_N^N &= 1, & W_N^{j+N} &= W_N^j \\ W_N^j &= W_N^{j \bmod N} & 0 \leq j < N \\ X(j) &= X(j \bmod N) \\ A(n) &= A(n \bmod N) \end{aligned}$$

Orthogonality:

$$\sum_{j=0}^{N-1} W_N^{nj} W_N^{-mj} = \begin{cases} N & \text{if } n = m \bmod N \\ 0 & \text{otherwise.} \end{cases}$$

Fig. 1. Definition of discrete Fourier transform.

③ S. 28 - Fig. 1

Mapping of Indices:

Assume $N = r \times s$,

$$\begin{aligned} j &\rightarrow (j_1, j_2) & j_1 &= 0, 1, \dots, r-1 \\ i &\rightarrow (i_1, i_2) & i_1 &= 0, 1, \dots, s-1 \end{aligned}$$

$$\begin{aligned} n &\rightarrow (n_1, n_2) & n_1 &= 0, 1, \dots, s-1 \\ n &= n_1 s + n_2 & n_2 &= 0, 1, \dots, r-1 \end{aligned}$$

$$\begin{aligned} W_N^{jn} &= W_N^{j_1 n_1 r + j_2 n_2 s} = W_N^{j_1 n_1 r} \times W_N^{j_2 n_2 s} \\ &= 1 \times W_N^{j_1 n_1 r} = W_N^{j_1 n_1} \times W_N^{j_2 n_2} \\ X(j_1, j_2) &= \sum_{n_1=0}^{s-1} \sum_{n_2=0}^{r-1} A(n_1, n_2) W_N^{j_1 n_1} W_N^{j_2 n_2} \\ &= \sum_{n_1=0}^{s-1} A_1(j_1, n_1) W_N^{j_1 n_1} \end{aligned}$$

$$X(j_1, j_2) = \sum_{n_1=0}^{s-1} A_1(j_1, n_1) W_N^{j_1 n_1}$$

Fig. 2. Factorization into subseries.

Number of operations (multiply-adds) for N -point Fourier analysis is:

Number of operations = N^2

By factoring, $N = r \times s$,

Number of operations = $r^2 s + s^2 r + N \times (r+s)$.

By factoring $N = r_1 \times r_2 \times \dots \times r_m$:

Number of operations = $N \times (r_1 + r_2 + \dots + r_m)$.

For minimum, $r_1 = \dots = r_m = r$, $N = r^m$, i.e.,

$m = \log_2 N$:

Number of operations = $N \times r \times \log_2 N = N \times \log_2 N \times \frac{r}{\log_2 r}$

r	r log ₂ r
2	2.00
3	1.86
4	2.00
5	2.15
6	2.21
7	2.49
8	2.67
10	2.82

Fig. 3. Minimization of number of operations.

③ S. 28 - Fig. 2, 3

$$X_n = \sum_{l=-\infty}^{\infty} c_n + lN. \quad (11.4)$$

In the calculation (11.3) we have used the fact that the series (11.1) converges absolutely. Combining (11.2) and (11.3), we obtain

$$x_k := x(t_k) = \sum_{n=-N/2}^{N/2-1} X_n e^{i2\pi k n / N}. \quad (11.5)$$

The formula (11.5) can be viewed as an inverse discrete Fourier transform, and it appeared quite naturally in the discretization of a continuous periodic signal. Moreover, the formula (11.4) becomes the main property of this approach. Since

$$\sum_{n=-N/2}^{N/2-1} e^{i(k-m)2\pi n / N} = \begin{cases} 0, & k-m \neq 0, \pm N, \pm 2N, \dots \\ N, & k-m = 0, \pm N, \pm 2N, \dots \end{cases} \quad (11.6)$$

we solve the linear system (11.5) with respect to $X_n, n = -N/2, \dots, N/2-1$, and obtain

$$X_n = \frac{1}{N} \sum_{k=-N/2}^{N/2-1} x_k e^{-i2\pi k n / N}. \quad (11.7)$$

Exercise 11.1. Prove (11.6) and (11.7).

In fact, the formulas (11.5) and (11.7) give us the inverse and direct discrete Fourier transforms, respectively.

Definition 11.1. The sequence $\{X_n\}_{n=-N/2}^{N/2-1}$ of complex numbers is called the *discrete Fourier transform (DFT)* of the sequence $\{x_k\}_{k=-N/2}^{N/2-1}$ if for each $n = -N/2, \dots, N/2-1$ we have

$$X_n = \frac{1}{N} \sum_{k=-N/2}^{N/2-1} x_k e^{-i2\pi k n / N}. \quad (11.8)$$

We use the symbol $\mathcal{F}$ for the DFT and write

$$X_n = \mathcal{F}\{x_k\}_n$$

or simply $X = \mathcal{F}(Y)$.

Definition 11.2. The sequence $\{Z_k\}_{k=-N/2}^{N/2-1}$ of complex numbers is said to be the *inverse discrete Fourier transform (IDFT)* of the sequence $\{X_n\}_{n=-N/2}^{N/2-1}$ if for each $k = -N/2, \dots, N/2-1$ we have

① s. 78 | Ch. 11

① s. 79 | Ch. 11

Definition 11.7. The *convolution* of discrete sequences $X = \{X_n\}_{n=-N/2}^{N/2-1}$ and $Y = \{Y_n\}_{n=-N/2}^{N/2-1}$ is defined as the sequence whose elements are given by

$$(X * Y)_k = \sum_{l=-N/2}^{N/2-1} X_l Y_{k-l}, \quad (11.11)$$

where X_n and Y_n satisfy the periodicity condition (11.10).

③ s. 80 | Ch. 11

→ Verwendung DFT - FFT ① s. 85 - 91 (Analogie Sampling etc. noch interessant...)

1.6 FFT

```

SUBROUTINE FFT(A,M)
  COMPLEX A(1024),U,M,T
  N= 2**M
  NV2 = N/2
  NA=1 = N-1
  J=1
  DO 7 I=1,NM1
    IF(1,GE,J) GO TO 5
    T = A(J)
    A(J) = A(I)
    A(I) = T
  5 K=NV2
  6 IF(K,GE,J) GO TO 7
    J = J-K
    K=K/2
    GO TO 6
  7 J=J+K
    PI = 3.14159265358979
    DO 20 L=1,M
      LE = 2**L
      LE1 = LE/2
      U = (1.0,0.0)
      W=CMPLX(COS(PI/LE1),SIN(PI/LE1))
      DO 20 J=1,LE1
        DO 10 I=J,N,LE
          IP = I+LE1
          T=A(IP)+U
          A(IP)=A(I)-T
          10 A(I)=A(IP)+T
        20 U=U*W
      RETURN
    END

```

Fig. 5. Program for computing DFT by FFT method.

③ s. 29 | Fig. 5 ← Text devor auch gut gut?

noch bessere Def für FFT

2. Konvergenzresultate

2.1 Mean-square convergence

Theorem 1.1 Suppose f is integrable on the circle. Then

$$\frac{1}{2\pi} \int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2 d\theta \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

As we remarked earlier, the key concept involved is that of orthogonality. The correct setting for orthogonality is in a vector space equipped with an inner product.

konkrete L^2 -Konvergenz

② s. 70 | Ch. 3

2.2 Hilbertraumstruktur - warum Orthogonalität der rechte Begriff ist

1.1 Vector spaces and inner products

We now review the definitions of a vector space over $\mathbb{R}$ or $\mathbb{C}$, an inner product, and its associated norm. In addition to the familiar finite-dimensional vector spaces $\mathbb{R}^d$ and $\mathbb{C}^d$, we also examine two infinite-dimensional examples which play a central role in the proof of Theorem 1.1.

Preliminaries on vector spaces

A vector space V over the real numbers $\mathbb{R}$ is a set whose elements may be "added" together, and "multiplied" by scalars. More precisely, we may associate to any pair $X, Y \in V$ an element in V called their sum and denoted by $X + Y$. We require that this addition respects the usual laws

$$Z_k = \sum_{n=-N/2}^{N/2-1} X_n e^{i2\pi k n / N}. \quad (11.9)$$

We use the symbol $\mathcal{F}^{-1}$ for the IDFT and write

$$Z_k = \mathcal{F}^{-1}\{X_n\}_k$$

or simply $Z = \mathcal{F}^{-1}(X)$.

The properties of the DFT and IDFT are collected in the following lemmas.

Lemma 11.3. The following equalities hold:

(1) $\mathcal{F}^{-1}(\mathcal{F}(Y)) = Y$;

(2) $\mathcal{F}(\mathcal{F}^{-1}(X)) = X$;

(3)

$$\sum_{k=-N/2}^{N/2-1} \mathcal{F}(X_n)_k \overline{\mathcal{F}(Y_n)_k} = \frac{1}{N} \sum_{n=-N/2}^{N/2-1} X_n \overline{Y_n}.$$

Proof. Using (11.6), (11.8), and (11.9), we have

$$\begin{aligned} \mathcal{F}^{-1}(\mathcal{F}(Y))_k &= \sum_{n=-N/2}^{N/2-1} \mathcal{F}(Y)_n e^{i2\pi k n / N} = \frac{1}{N} \sum_{n=-N/2}^{N/2-1} \left(\sum_{l=-N/2}^{N/2-1} Y_l e^{-i2\pi l n / N} \right) e^{i2\pi k n / N} \\ &= \frac{1}{N} \sum_{l=-N/2}^{N/2-1} Y_l \left(\sum_{n=-N/2}^{N/2-1} e^{i2\pi n (k-l) / N} \right) = \frac{1}{N} Y_l N = Y_l. \end{aligned}$$

This proves part (1). Part (2) can be proved in the same manner. $\square$

Exercise 11.2. Prove part (3) of Lemma 11.3.

Corollary 11.4 (Parseval's equality).

$$\frac{1}{N} \sum_{n=-N/2}^{N/2-1} |X_n|^2 = \sum_{k=-N/2}^{N/2-1} |\mathcal{F}(X_n)_k|^2.$$

Remark 11.5. Due to the periodicity of the complex exponential, we may extend the values of $X_n, n = -N/2, \dots, N/2-1$ periodically to any integer by

$$X_{n+N} = X_n, \quad l = 0, \pm 1, \pm 2, \dots \quad (11.10)$$

Corollary 11.6. For a sequence $X = \{X_n\}_{n=-N/2}^{N/2-1}$ we define

$$X_{\text{ev}} = \{X_{N-n}\}_{n=-N/2}^{N/2-1}.$$

of arithmetic, such as commutativity $X + Y = Y + X$, and associativity $X + (Y + Z) = (X + Y) + Z$, etc. Also, given any $X \in V$ and real number λ , we assign an element $\lambda X \in V$ called the product of X by λ . This scalar multiplication must satisfy the standard properties, for instance $\lambda_1(\lambda_2 X) = (\lambda_1 \lambda_2)X$ and $\lambda(X + Y) = \lambda X + \lambda Y$. We may instead allow scalar multiplication by numbers in $\mathbb{C}$; we then say that V is a vector space over the complex numbers.

② S. 10 | Ch. 3

For example, the set $\mathbb{R}^d$ of d -tuples of real numbers $(x_1, x_2, \dots, x_d)$ is a vector space over the reals. Addition is defined componentwise by

$$(x_1, \dots, x_d) + (y_1, \dots, y_d) = (x_1 + y_1, \dots, x_d + y_d),$$

and so is multiplication by a scalar $\lambda \in \mathbb{R}$:

$$\lambda(x_1, \dots, x_d) = (\lambda x_1, \dots, \lambda x_d).$$

Similarly, the space $\mathbb{C}^d$ (the complex version of the previous example) is the set of d -tuples of complex numbers $(z_1, z_2, \dots, z_d)$. It is a vector space over $\mathbb{C}$ with addition defined componentwise by

$$(z_1, \dots, z_d) + (w_1, \dots, w_d) = (z_1 + w_1, \dots, z_d + w_d).$$

Multiplication by scalars $\lambda \in \mathbb{C}$ is given by

$$\lambda(z_1, \dots, z_d) = (\lambda z_1, \dots, \lambda z_d).$$

An **inner product** on a vector space V over $\mathbb{R}$ associates to any pair X, Y of elements in V a real number which we denote by (X, Y) . In particular, the inner product must be symmetric $(X, Y) = (Y, X)$ and linear in both variables; that is,

$$(\alpha X + \beta Y, Z) = \alpha(X, Z) + \beta(Y, Z)$$

whenever $\alpha, \beta \in \mathbb{R}$ and $X, Y, Z \in V$. Also, we require that the inner product be positive-definite, that is, $(X, X) \geq 0$ for all X in V . In particular, given an inner product $(\cdot, \cdot)$ we may define the norm of X by

$$\|X\| = (X, X)^{1/2}.$$

If in addition $\|X\| = 0$ implies $X = 0$, we say that the inner product is strictly positive-definite.

For example, the space $\mathbb{R}^d$ is equipped with a (strictly positive-definite) inner product defined by

$$(X, Y) = x_1 y_1 + \dots + x_d y_d$$

when $X = (x_1, \dots, x_d)$ and $Y = (y_1, \dots, y_d)$. Then

$$\|X\| = (X, X)^{1/2} = \sqrt{x_1^2 + \dots + x_d^2},$$

was hat das mit
Hilbertraum zu tun?
→ wenn brauche ich das B?

② S. 71 | Ch. 3

23. Konvergent auf $\mathbb{C}^1$

Theorem 5.2 If f is integrable on the circle, then the Fourier series of f is Cesàro summable to f at every point of continuity of f .

Moreover, if f is continuous on the circle, then the Fourier series of f is uniformly Cesàro summable to f .

② S. 53 | Ch. 2 (S. 51-54)

2.4. Konvergent auf $\mathbb{L}^2$

Theorem 2.1 Let f be an integrable function on the circle which is differentiable at a point θ_0 . Then $S_N(f)(\theta_0) \rightarrow f(\theta_0)$ as N tends to infinity.

② S. 81 | Ch. 3 (Erläuterung + Proof = S. 86) → Differenzierbarkeit result..

2.5. Warum man punktweise Konvergent vermeidet

→ Gegenbeispiel: siehe Pl mit divergierendes Fourier-Reihe
(Symmetry-breaking Prinzip)

2.2 A continuous function with diverging Fourier series

We now turn our attention to an example of a continuous periodic function whose Fourier series diverges at a point. Thus, Theorem 2.1 fails if the differentiability assumption is replaced by the weaker assumption of continuity. Our counter-example shows that this hypothesis which had appeared plausible, is in fact false; moreover, its construction also illuminates an important principle of the theory.

The principle that is involved here will be referred to as “symmetry-breaking.”¹ The symmetry that we have in mind is the symmetry between the frequencies $e^{in\theta}$ and $e^{-in\theta}$ which appear in the Fourier expansion of a function. For example, the partial sum operator S_N is defined in a way that reflects this symmetry. Also, the Dirichlet, Fejér, and Poisson kernels are symmetric in this sense. When we break the symmetry, that is, when we split the Fourier series $\sum_{n=-\infty}^{\infty} a_n e^{in\theta}$ into the two pieces $\sum_{n \geq 0} a_n e^{in\theta}$ and $\sum_{n < 0} a_n e^{in\theta}$, we introduce new and far-reaching phenomena.

We give a simple example. Start with the sawtooth function f which is odd in θ and which equals $i(\pi - \theta)$ when $0 < \theta < \pi$. Then, by Exercise 8 in Chapter 2, we know that

$$(4) \quad f(\theta) \sim \sum_{n \neq 0} \frac{e^{in\theta}}{n}.$$

Consider now the result of breaking the symmetry and the resulting series

$$\sum_{n=0}^{\infty} \frac{e^{in\theta}}{n}.$$

Then, unlike (4), the above is no longer the Fourier series of a Riemann integrable function. Indeed, suppose it were the Fourier series of an

integrable function, say f , where in particular f is bounded. Using the Abel means, we then have

$$|A_r(\tilde{f})(\theta)| \leq \sum_{n=1}^{\infty} \frac{r^n}{n},$$

which tends to infinity as r tends to 1, because $\sum 1/n$ diverges. This gives the desired contradiction since

$$|A_r(\tilde{f})(0)| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |\tilde{f}(\theta)| P_r(\theta) d\theta \leq \sup_{\theta} |\tilde{f}(\theta)|,$$

where $P_r(\theta)$ denotes the Poisson kernel discussed in the previous chapter.

The sawtooth function is the object from which we will fashion our counter-example. We proceed as follows. For each $N \geq 1$ we define the following two functions on $[-\pi, \pi]$,

$$f_N(\theta) = \sum_{1 \leq |n| \leq N} \frac{e^{in\theta}}{n} \quad \text{and} \quad \tilde{f}_N(\theta) = \sum_{-N \leq n \leq -1} \frac{e^{in\theta}}{n}.$$

We contend that:

- (i) $|\tilde{f}_N(0)| \geq c \log N$.
- (ii) $f_N(\theta)$ is uniformly bounded in N and θ .

The first statement is a consequence of the fact that $\sum_{n=1}^N 1/n \geq \log N$, which is easily established (see also Figure 2):

$$\sum_{n=1}^N \frac{1}{n} \geq \sum_{n=1}^{N-1} \int_n^{n+1} \frac{dx}{x} = \int_1^N \frac{dx}{x} = \log N.$$

To prove (ii), we argue in the same spirit as in the proof of Tauber's theorem, which says that if the series $\sum c_n$ is Abel summable to s and $c_n = o(1/n)$, then $\sum c_n$ actually converges to s (see Exercise 14 in Chapter 2). In fact, the proof of Tauber's theorem is quite similar to that of the lemma below.

Lemma 2.3 Suppose that the Abel means $A_r = \sum_{n=1}^{\infty} r^n c_n$ of the series $\sum_{n=1}^{\infty} c_n$ are bounded as r tends to 1 (with $r < 1$). If $c_n = O(1/n)$, then the partial sums $S_N = \sum_{n=1}^N c_n$ are bounded.

¹We have borrowed this terminology from physics, where it is used in a very different context.

② S. 83 | Ch. 3

② S. 84 | Ch. 3

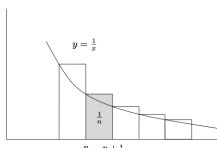


Figure 2. Comparing a sum with an integral

Proof. Let $r = 1 - 1/N$ and choose M so that $n|c_n| \leq M$. We estimate the difference

$$S_N - A_r = \sum_{n=1}^N (c_n - r^n c_n) - \sum_{n=N+1}^{\infty} r^n c_n$$

as follows:

$$\begin{aligned} |S_N - A_r| &\leq \sum_{n=1}^N |c_n| (1 - r^n) + \sum_{n=N+1}^{\infty} r^n |c_n| \\ &\leq M \sum_{n=1}^N (1 - r) + \frac{M}{N} \sum_{n=N+1}^{\infty} r^n \\ &\leq MN(1 - r) + \frac{M}{N} \frac{1}{1 - r} \\ &= 2M, \end{aligned}$$

where we have used the simple observation that

$$1 - r^n = (1 - r)(1 + r + \dots + r^{n-1}) \leq n(1 - r).$$

So we see that if M satisfies both $|A_r| \leq M$ and $n|c_n| \leq M$, then $|S_N| \leq 3M$.

We apply the lemma to the series

$$\sum_{n \neq 0} \frac{e^{in\theta}}{n},$$

② 5.85 | Ch. 3

2.6 Dirichlet - Kern

Rechen für Konvergenztheorie - wirksam

Chapter 10

Dirichlet Kernel. Pointwise and Uniform Convergence.

The material of this chapter forms a central part of the theory of trigonometric Fourier series. In this chapter we will answer the following question: to what value does a trigonometric Fourier series converge?

The *Dirichlet kernel* $D_N(x)$, which is defined by the symmetric finite trigonometric sum

$$D_N(x) := \sum_{|n| \leq N} e^{inx}, \quad (10.1)$$

plays a key role in this chapter. If $x \in [-\pi, \pi] \setminus \{0\}$, then $D_N(x)$ from (10.1) can be recalculated as follows. Using Euler's formula, we have

$$\begin{aligned} D_N(x) &= \sum_{n=-N}^N e^{inx} = e^{-iNx} \sum_{n=-N}^N e^{i(n+N)x} = e^{-iNx} \sum_{k=0}^{2N} e^{ikx} \\ &= e^{-iNx} \frac{1 - e^{i(2N+1)x}}{1 - e^{ix}} = e^{-iNx} \frac{1 - e^{i(2N+1)x}}{1 - e^{ix}} \\ &= \frac{e^{i(N+1/2)x} - e^{-i(N+1/2)x}}{e^{ix/2} - e^{-ix/2}} = \frac{\sin(N + 1/2)x}{\sin x/2}. \end{aligned}$$

Thus, the Dirichlet kernel equals

$$D_N(x) = \frac{\sin(N + 1/2)x}{\sin x/2}, \quad x \neq 0. \quad (10.2)$$

For $x = 0$ we have

$$D_N(0) = 2N + 1 = \lim_{x \rightarrow 0} D_N(x),$$

so that (10.2) holds for all $x \in [-\pi, \pi]$.

② 5.59 (ext. auch 2.69) | Ch. 10

→ ② 5.63-67 Bed für punktweise / gleichmäßige Konvergenz via Dirichlet-Kern

3. Denoising mit FT

3.1. Convolutions

3 Convolutions

The notion of convolution of two functions plays a fundamental role in Fourier analysis: it appears naturally in the context of Fourier series but also serves more generally in the analysis of functions in other settings. Given two 2π -periodic integrable functions f and g on $\mathbb{R}$, we define their *convolution* $f * g$ on $[-\pi, \pi]$ by

$$(f * g)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y)g(x-y) dy.$$

The above integral makes sense for each x , since the product of two integrable functions is again integrable. Also, since the functions are periodic, we can change variables to see that

$$(f * g)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-y)g(y) dy.$$

Loosely speaking, convolutions correspond to "weighted averages." For instance, if $g = 1$ in (2), then $f * g$ is constant and equal to $\frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) dy$, which we may interpret as the average value of f on the circle. Also, the convolution $(f * g)(x)$ plays a role similar to, and in some sense replaces, the pointwise product $f(x)g(x)$ of the two functions f and g .

In the context of this chapter, our interest in convolutions originates from the fact that the partial sums of the Fourier series of f can be expressed as follows:

$$\begin{aligned} S_N(f)(x) &= \sum_{n=-N}^N f(n)e^{inx} \\ &= \sum_{n=-N}^N \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(y)e^{-iny} dy \right) e^{inx} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \left(\sum_{n=-N}^N e^{in(x-y)} \right) dy \\ &= (f * D_N)(x), \end{aligned}$$

② 5.44 | Ch. 2

where D_N is the N^{th} Dirichlet kernel (see Example 4) given by

$$D_N(x) = \sum_{n=-N}^N e^{inx}.$$

So we observe that the problem of understanding $S_N(f)$ reduces to the understanding of the convolution $f * D_N$.

We begin by gathering some of the main properties of convolutions.

Proposition 3.1 Suppose that f , g , and h are 2π -periodic integrable functions. Then:

- (i) $f * (g + h) = (f * g) + (f * h)$.
- (ii) $(cf) * g = c(f * g) = f * (cg)$ for any $c \in \mathbb{C}$.
- (iii) $f * g = g * f$.
- (iv) $(f * g) * h = f * (g * h)$.
- (v) $f * g$ is continuous.
- (vi) $\widehat{f * g}(n) = \widehat{f}(n)\widehat{g}(n)$.

The first four points describe the algebraic properties of convolutions: linearity, commutativity, and associativity. Property (v) exhibits an important principle: the convolution of $f * g$ is "more regular" than f or g . Here, $f * g$ is continuous while f and g are merely (Riemann) integrable. Finally, (vi) is key in the study of Fourier series. In general, the Fourier coefficients of the product fg are not the product of the Fourier coefficients of f and g . However, (vi) says that this relation holds if we replace the product of the two functions f and g by their convolution $f * g$.

② S. 45 / Ch. 2

3.2. Faltungssatz für OFT

Convolution:

$$Z_j = \sum_{k=0}^{N-1} X_k Y_{j-k}$$

i.e.,

$$\begin{pmatrix} Z_0 \\ Z_1 \\ Z_2 \\ \vdots \\ Z_{N-1} \end{pmatrix} = \begin{pmatrix} Y_0 & Y_{N-1} & Y_{N-2} & \cdots & Y_1 \\ Y_1 & Y_0 & Y_{N-1} & \cdots & Y_2 \\ Y_2 & Y_1 & Y_0 & \cdots & Y_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Y_{N-1} & Y_{N-2} & Y_{N-3} & \cdots & Y_0 \end{pmatrix} \begin{pmatrix} X_0 \\ X_1 \\ X_2 \\ \vdots \\ X_{N-1} \end{pmatrix}$$

Theorem: If

$$\begin{aligned} X_j &\leftrightarrow A_n \\ Y_j &\leftrightarrow B_n \\ Z_j &\leftrightarrow C_n \end{aligned}$$

then

$$C_n = N \cdot A_n \cdot B_n$$

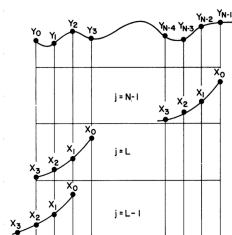
Fig. 13. Convolution theorem.

③ S. 32 / Fig 13

→ Erklärung ③ S. 32 / 33
we faltung im Frequenzbereich zur Multiplikation und
→ Grundlage für spektrales Filtern

3.3. digitales Filtern

$$Z_j = \sum_{k=0}^L X_k Y_{j-k}$$



Number of Operations per Filter Output:
By Sum of Products:

$$\text{By Fourier Transform Methods: } \frac{\text{No. Operations}}{\text{No. Filter Outputs}} = \frac{2N \log N}{N-1}$$

Fig. 15. Digital filtering.

— konkrete Anwendung des Faltungssatzes

③ S. 33 — Fig 15

→ operationenanzahl $2N \log N$ vs direkte Methode ...

→ hier wäre konkrete Anwendung FKS gut!

4. Limitations of the FT

4.1 Heisenbergsche Unschärferelation

Grundlage des: FT ist global — Argumente!

4 The Heisenberg uncertainty principle

The mathematical thrust of the principle can be formulated in terms of a relation between a function and its Fourier transform. The basic underlying law, formulated in its vaguest and most general form, states that a function and its Fourier transform cannot both be essentially localized. Somewhat more precisely, if the "preponderance" of the mass of a function is concentrated in an interval of length L , then the preponderance of the mass of its Fourier transform cannot lie in an interval of length essentially smaller than L^{-1} . The exact statement is as follows.

Theorem 4.1 Suppose ψ is a function in $S(\mathbb{R})$ which satisfies the normalizing condition $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$. Then

$$\left(\int_{-\infty}^{\infty} x^2 |\psi(x)|^2 dx \right) \left(\int_{-\infty}^{\infty} \xi^2 |\hat{\psi}(\xi)|^2 d\xi \right) \geq \frac{1}{16\pi^2},$$

and equality holds if and only if $\psi(x) = Ae^{-Bx^2}$ where $B > 0$ and $|A|^2 = \sqrt{2B/\pi}$.

In fact, we have

$$\left(\int_{-\infty}^{\infty} (x - x_0)^2 |\psi(x)|^2 dx \right) \left(\int_{-\infty}^{\infty} (\xi - \xi_0)^2 |\hat{\psi}(\xi)|^2 d\xi \right) \geq \frac{1}{16\pi^2}$$

for every $x_0, \xi_0 \in \mathbb{R}$.

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The precise assertion contained in Theorem 4.1 first came to light in the study of quantum mechanics. It arose when one considered the extent to which one could simultaneously locate the position and momentum of a particle. Assuming we are dealing with (say) an electron that travels along the real line, then according to the laws of physics, matters are governed by a "state function" ψ , which we can assume to be in $S(\mathbb{R})$, and which is normalized according to the requirement that

$$(11) \quad \int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1.$$

The position of the particle is then determined not as a definite point x ; instead its probable location is given by the rules of quantum mechanics as follows:

- The probability that the particle is located in the interval (a, b) is $\int_a^b |\psi(x)|^2 dx$.

According to this law we can calculate the probable location of the particle with the aid of ψ : in fact, there may be only a small probability that the particle is located in a given interval (a', b') , but nevertheless it is somewhere on the real line since $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$.

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eine flach und ihre FT können nicht beide unendlich lokalisiert sein...